

Executive Summary

Salisfort Motors

HR Dataset Analysis

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*full code and analysis process can be found on GitHub

Salifort Motors is experiencing challenges with employee retention and has engaged our data analytics services to address this issue. The HR department has collected employee data but requires **data-driven insights** to inform initiatives aimed at improving employee satisfaction and reducing leaving rates.

Out of 11,991 employees

16.6%

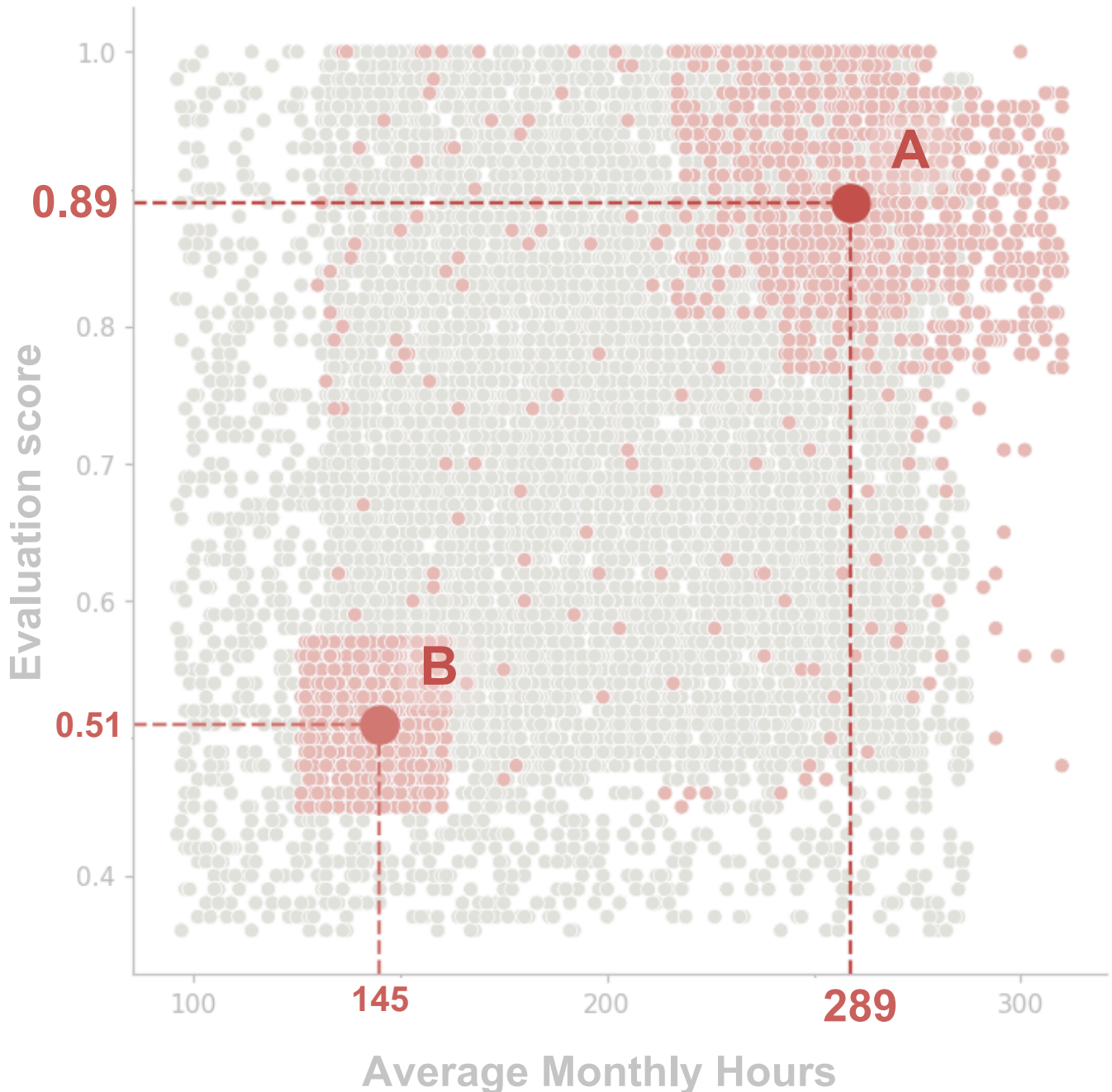
have left the company

This report details findings of the provided dataset, focusing on identifying the factors most likely to contribute to employee departures. The goal is to develop a **predictive model** for employee attrition, enabling Salifort Motors to proactively identify at-risk employees and implement targeted retention strategies, ultimately reducing the costs and disruptions associated with employee turnover.

Skilled & Overworked are leaving

Less efficient workers are getting fired

Monthly Hours & Latest Evaluation
of those who **Left** and Stayed

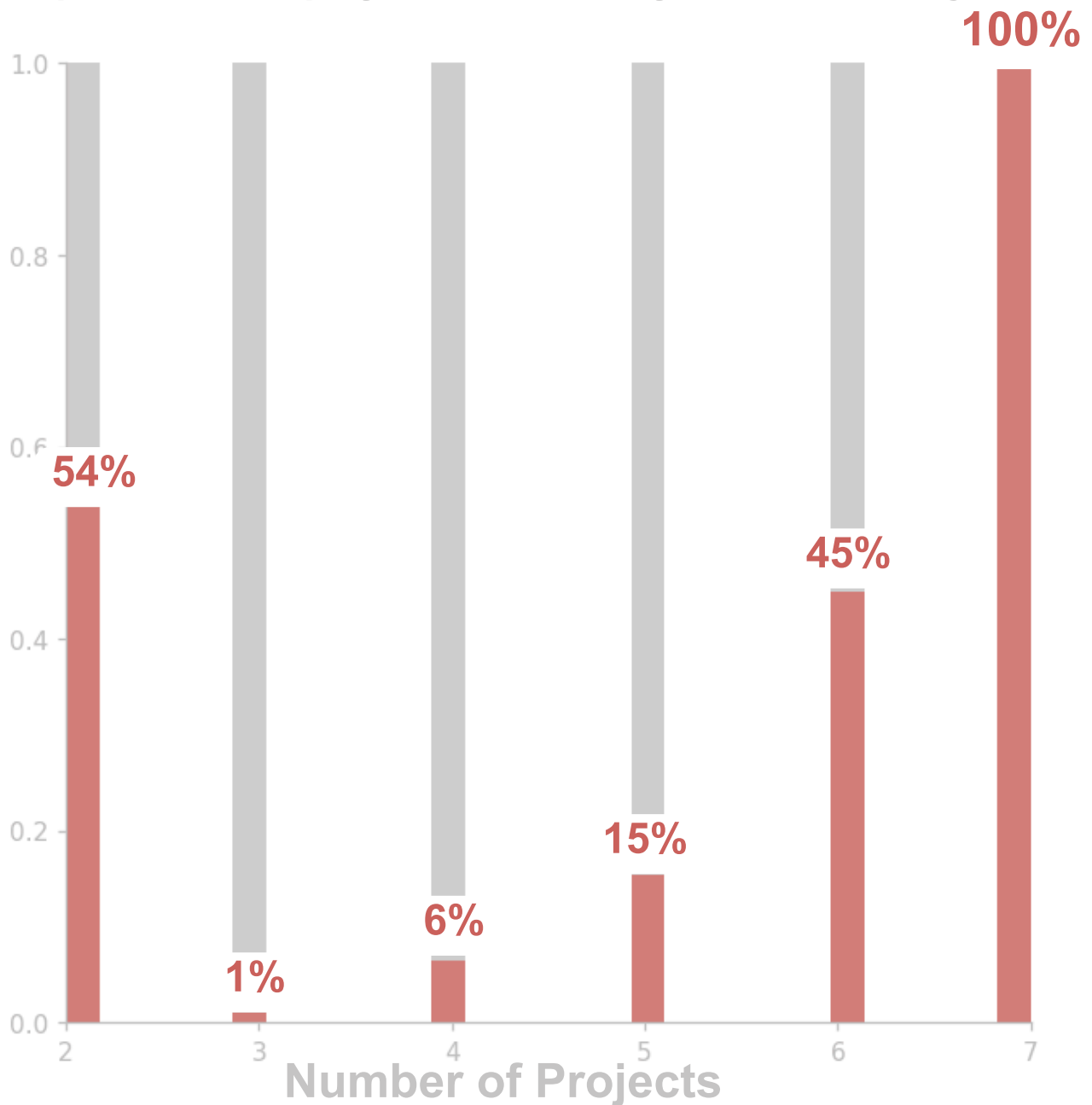


The data reveals two distinct groups of departing employees: (A) high-performing, overworked individuals, and (B) lower-performing employees working fewer hours (avg. 145). Group B may have been fired or reduced hours due to pre-planned departures. Group A likely resigned and potentially made significant contributions to the company.

Limit the maximum number of projects to 4

Never let anybody work on 7 projects at the same time

Proportion of Employees Who **Left** by Number of Projects

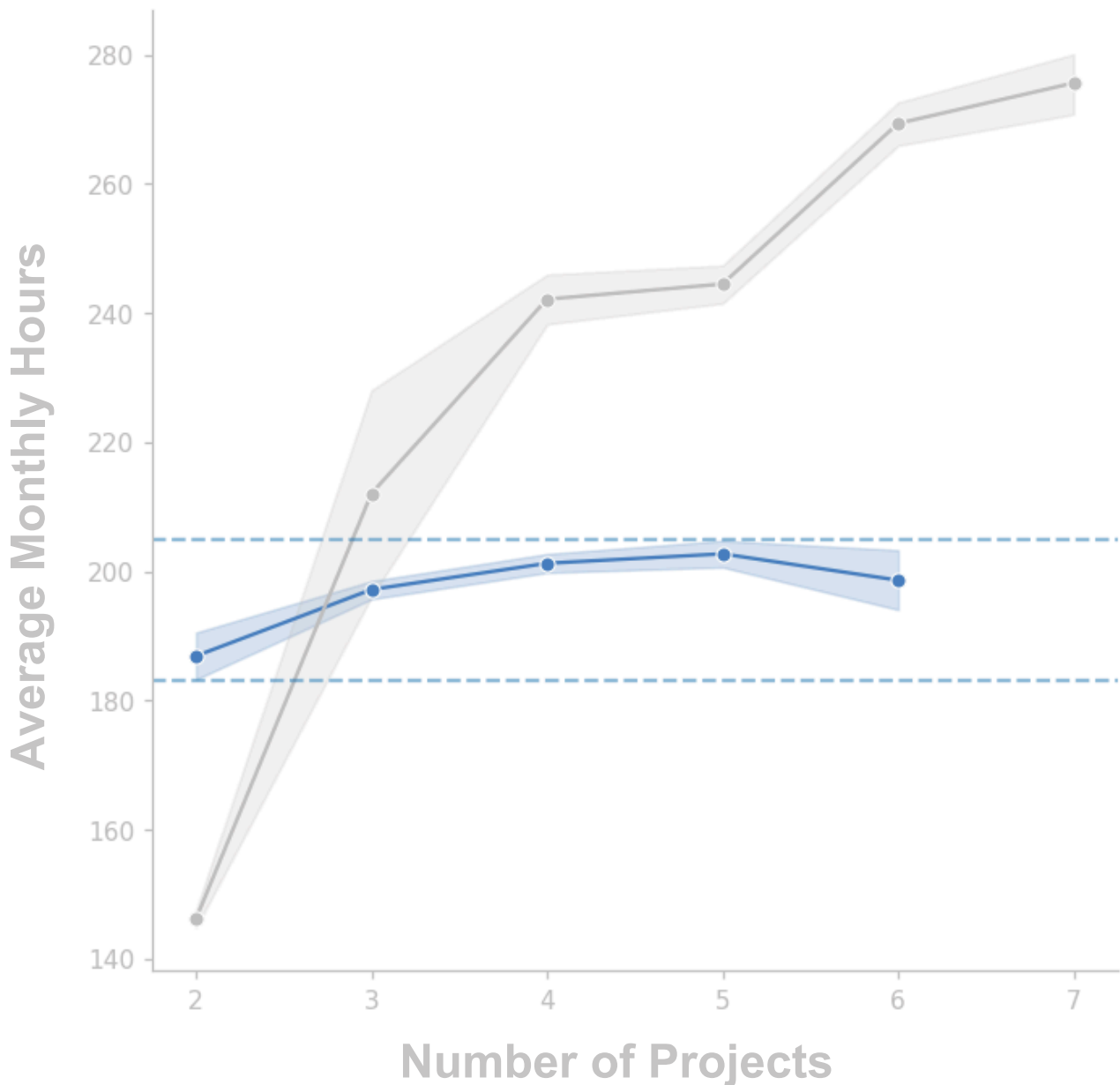


Proportion of employees who left with a manageable workload (3-4 projects) is significantly lower. However, as the number of projects increases, the proportion of employees leaving rises dramatically, reaching 100% for those with 7 projects. This data strongly suggests that limiting the number of projects per employee, ideally to a maximum of 4, is critical for improving employee retention.

Those who work 180-205 hours per month stay

Encourage a work-life balance that aligns with the 180-205 hour work month sweet spot

of Projects & Mean Monthly Hours of those who Left and Stayed



There is a correlation between work hours and employee retention, highlighting an optimal range of 180-205 hours per month. Employees who left consistently worked more hours as their project load increased, while those who stayed maintained hours closer to this ideal range. Managing workload to prevent excessive work hours is crucial for retention.

The ML Model can predict

93%

**of those who might be
at risk of leaving**

To evaluate predictive model, analysis focused on influential employee characteristics for predicting departures and assessed predictive accuracy. Model identified number of projects, work hours, and recent performance reviews as some of the strongest indicators of likely departures. Tool effectively identifies 93 out of 100 employees at risk of leaving and demonstrates 97 out of 100 times accuracy when flagging a potential departure.

Summary

Key Findings:

- High workload (number of projects) and excessive work hours are major factors in employees leaving.

Recommendations:

- Limit the number of projects assigned to each employee to a maximum of 4.
- Promote a work-life balance that aligns with a 180-205 hour work month.
- Utilize the machine learning model to proactively identify at-risk employees and provide targeted interventions.